

Erdős #1094: the least prime factor of $\binom{n}{k}$ as a domination threshold

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Abstract

#1094 asserts that for $n \geq 2k$ the least prime factor of $\binom{n}{k}$ is at most $\max(n/k, k)$, with only finitely many exceptions. We give the exact ledger form of $\text{lpf} \binom{n}{k}$ and prove that the primes above $n/2$ never contribute to it. The main result is that *every exception has $\binom{n}{k}$ squarefree*: since $\max(n/k, k) \geq \sqrt{n}$, an exception forces digit domination at every rail $p \leq \sqrt{n}$, hence every prime factor of $\binom{n}{k}$ exceeds $\sqrt{n}$ and Legendre's formula leaves each exponent at most 1. Consequently #1094 follows from finiteness of the squarefree binomial coefficients, which places it beside #175 and #378 rather than among least-prime-factor estimates. A census isolates Selfridge's $\binom{62}{6}$ as the unique exception in his conjectured range. #1094 itself is not proved.

1 Ledger form

Write $k \preceq_p n$ for base- p digit domination. By Kummer, $p \mid \binom{n}{k}$ iff the base- p addition $k + (n - k)$ carries, iff $k \not\preceq_p n$. Hence

Proposition 1. $\text{lpf} \binom{n}{k} = \min\{p : k \not\preceq_p n\}$, *computable without evaluating any binomial coefficient.*

Verified: agrees with direct computation of the least prime factor for $2 \leq k \leq 8$, $n < 400$ — no mismatch.

Lemma 2 (large rails are irrelevant). *Let $1 \leq k \leq n/2$ and $p > n/2$ prime. Then $v_p \binom{n}{k} = 1$ if $n - k < p \leq n$ and 0 if $n/2 < p \leq n - k$. In particular the least prime factor is never a prime exceeding $n/2$ unless $\binom{n}{k}$ has no smaller prime factor at all.*

Proof. For $m \leq n$ and $p > n/2$ we have $p^2 > n \geq m$, so $v_p(m!) = \lfloor m/p \rfloor$; thus $v_p \binom{n}{k} = 1 - 0 - \lfloor (n - k)/p \rfloor$ since $k \leq n/2 < p$. $\square$

So lpf is governed entirely by the small rails, in contrast to $P \binom{n}{k}$ (#683) and $\omega \binom{n}{k}$ (#685), to which the window $(n - k, n]$ contributes directly. The threshold below which $\binom{n}{k}$ is guaranteed a prime factor $\leq k$ is the subject of #1095, whose criterion $k \preceq_p n$ for all $p \leq k$ is exactly the negation of Proposition 1 restricted to $[2, k]$.

2 Every exception has a squarefree binomial coefficient

Lemma 3. *For any real $B \geq 2$, $\text{lpf} \binom{n}{k} > B$ if and only if $k \preceq_p n$ for every prime $p \leq B$.*

Proof. By Proposition 1, $\text{lpf} \binom{n}{k} = \min\{p : k \not\preceq_p n\}$, so this minimum exceeds B exactly when no prime $p \leq B$ fails domination. $\square$

Lemma 4. *For $1 \leq k \leq n$, $\max(n/k, k) \geq \sqrt{n}$.*

Proof. The two quantities have product n , so the larger is at least their geometric mean $\sqrt{n}$. $\square$

Theorem 5. *Let $n \geq 2k$ and suppose (n, k) is an exception to #1094, that is $\text{lpf} \binom{n}{k} > \max(n/k, k)$. Then every prime factor of $\binom{n}{k}$ exceeds $\sqrt{n}$, and consequently $\binom{n}{k}$ is squarefree.*

Proof. By Lemma 4 the hypothesis gives $\text{lpf} \binom{n}{k} > \sqrt{n}$, so by Lemma 3 we have $k \preceq_p n$ for every prime $p \leq \sqrt{n}$; equivalently no prime $p \leq \sqrt{n}$ divides $\binom{n}{k}$. Hence every prime factor p of $\binom{n}{k}$ satisfies $p > \sqrt{n}$, so $p^2 > n$ and Legendre's formula has a single term: $v_p \binom{n}{k} = \lfloor n/p \rfloor - \lfloor k/p \rfloor - \lfloor (n-k)/p \rfloor \leq 1$. Every exponent in the factorisation of $\binom{n}{k}$ is therefore at most 1. $\square$

Verified: all 14 exceptions with $n < 4000$, $k < 30$ — namely $(7, 3), (13, 4), (14, 4), (23, 5), (62, 6), (44, 8), (46, 10)$ — have $\binom{n}{k}$ squarefree and $\text{lpf} > \sqrt{n}$, with no exception to either step.

Corollary 6. *The exception set of #1094 injects into the set of pairs (n, k) with $n \geq 2k$ for which $\binom{n}{k}$ is squarefree and $\text{lpf} \binom{n}{k} > \sqrt{n}$. Hence #1094 follows from finiteness of that set.*

Proof. Immediate from Theorem 5, which supplies both properties. $\square$

The second condition cannot be dropped. For $k = 2$ the coefficient $\binom{n}{2} = n(n-1)/2$ is squarefree for infinitely many n (already $\binom{7}{2} = 21$), so the unrestricted set of pairs with $\binom{n}{k}$ squarefree is infinite and finiteness of *it* is not available as an input. The condition $\text{lpf} > \sqrt{n}$ is what carries the weight, and it is exactly what the scarcity results of Granville–Ramaré (used for #175, and by Aggarwal and Cambie for #378) do not supply: those bound how often $\binom{n}{k}$ is squarefree, not how often it is squarefree with every prime factor above $\sqrt{n}$.

At $k = 2$ there is in fact no exception at all: for even $n \geq 4$ the divisor $n/2 \geq 2$ of $\binom{n}{2}$ gives $\text{lpf} \binom{n}{2} \leq n/2$, and for odd $n \geq 5$ the divisor $(n-1)/2 \geq 2$ gives $\text{lpf} \binom{n}{2} < n/2$. So #1094 begins at $k = 3$, with $\binom{7}{3}$.

3 The bridge to #1095

The threshold $B = \max(n/k, k)$ splits the problem at the AM–GM point $k = \sqrt{n}$, and on one side of it #1094 is #1095.

Proposition 7 (sector I). *Suppose $k^2 \geq n$. Then $B = k$, and (n, k) is an exception to #1094 if and only if every prime factor of $\binom{n}{k}$ exceeds k — that is, if and only if n is a witness for k in the sense of #1095.*

Proof. $k^2 \geq n$ gives $k \geq n/k$, so $B = \max(n/k, k) = k$ and the defining inequality $\text{lpf} \binom{n}{k} > B$ reads $\text{lpf} \binom{n}{k} > k$. $\square$

Corollary 8. *Let $g(k)$ be the least $n \geq 2k$ all of whose $\binom{n}{k}$ has prime factors exceeding k . Then #1094 restricted to the range $k^2 \geq n$ is exactly the assertion that $g(k) > k^2$ for all sufficiently large k .*

Proof. By Proposition 7 a sector-I exception with parameter k exists iff some witness $n \leq k^2$ exists, iff $g(k) \leq k^2$. $\square$

Verified: the equivalence of Proposition 7 holds on all 2 140 509 pairs (n, k) with $n < 3000$, $2 \leq k \leq n/2$ and $k^2 \geq n$ — no mismatch; and the sector-I exceptions in that range are exactly thirteen, namely the fourteen of the census with $\binom{62}{6}$ removed, $6^2 = 36 < 62$ placing it in the complementary sector.

Measured: among $k \leq 20$ the values with $g(k) \leq k^2$ are $k = 3, 4, 5, 8, 10, 11, 16$, and extending the g -table one more step adds $k = 28$; these eight k carry witness sets $\{7\}$, $\{13, 14\}$, $\{23\}$, $\{44\}$, $\{46, 47, 74, 94, 95\}$, $\{47\}$, $\{241\}$, $\{284\}$ — thirteen pairs in all. The remaining exception, $\binom{62}{6}$, has $k^2 = 36 < 62$ and lies in the complementary sector, where $B = n/k > k$ makes the condition strictly stronger than #1095. So the fourteen split as $13 + 1$ across the AM–GM point, and the sector-I half of #1094 is a lower-bound question about #1095’s own function.

4 Census

Exceptions to $\text{lpf} \left(\binom{n}{k} \right) \leq \max(n/k, k)$ over $n < 3000$, $k < 25$: **13** in total, beginning

$$(n, k, \text{lpf}) = (7, 3, 5), (13, 4, 5), (14, 4, 7), (23, 5, 7), (62, 6, 19), (44, 8, 11).$$

Restricting to Selfridge’s range $n \geq k^2 - 1$ leaves **exactly one**: $(62, 6, 19)$, i.e. $\binom{62}{6}$ — precisely and only the exception he named. The ledger form therefore reproduces the unique counterexample of a published conjecture without being given it.

Remark 9 (register). **Proved:** Proposition 1 and Lemma 2, elementary from Kummer and Legendre; neither is claimed new. Lemmas 3, 4, Theorem 5 and Corollary 6: every exception to #1094 has $\binom{n}{k}$ squarefree, so #1094 follows from finiteness of the squarefree binomial coefficients. **Measured:** the 13 exceptions and the isolation of $\binom{62}{6}$ in the range $n \geq k^2 - 1$. **Not proved:** finiteness of the exception set, i.e. #1094 itself. **Falsifier:** a second exception with $n \geq k^2 - 1$ would refute Selfridge’s conjecture; the domination form is cheap enough to search $k \leq 100$, $n \leq 10^6$ routinely, and such a hit would be worth reporting on its own.